

Torsion Pendulum

Aim:-

1. To determine the rigidity modulus of the material of the wire of the torsion pendulum
2. To determine the moment of inertia of the disc of the torsion pendulum using two identical masses on the disc
3. To determine the moment of inertia of the disc of the torsion pendulum by direct method

Apparatus:

- The torsion pendulum
- A stop watch
- Two identical masses (0.2kg each)

Description:

Torsion pendulum consists of a heavy regular or irregular (usually cylindrical rod, disc or sphere) suspended at the end of a thin and long wire. The other end of the wire is clamped to a rigid support. When the body at the lower end is gently and slightly rotated by applying equal and opposite torque, the wire gets twisted. It is due to this twisting an internal restoring couple, due to elastic properties of the material of the wire, is produced. When the body is released this restoring couple brings the body back to equilibrium position. But due to its kinetic energy of rotation, body overshoots this position and the wire starts twisting in the opposite direction. A restoring couple is again set up in it arrests its motion and makes it return. The whole process is then repeated. Thus the body oscillates in the horizontal plane about the wire as axis. These oscillations are called torsional oscillations.

Principle:

Suppose the twist at any instant at the lower end of the wire be θ

Then the restoring couple = $-C\theta$

where C is the twisting couple per unit twist and minus sign shows that couple is directed opposite to the twist.

If I be the moment of inertia of the body about the axis of rotation and $d^2\theta/dt^2$ is angular acceleration at that instant,

then the deflecting couple = $I d^2\theta/dt^2$

According to Newton's III law, Deflecting couple = Restoring couple

$$\therefore I d^2\theta/dt^2 = -C\theta$$

$$d^2\theta/dt^2 = -\omega^2\theta \text{ with } \omega^2 = C/I$$

$$\therefore T = 2\pi\sqrt{\frac{I}{C}}$$

A- To find the Moment of Inertia by Direct Method

For circular disc of mass M and radius R , $I = \frac{MR^2}{2}$

B- To find the rigidity modulus

If r = radius of the wire, L = length of the wire and G = rigidity modulus of the material of the wire,

$$C = \frac{\pi G r^4}{2 L}$$

$$\therefore G = \frac{8 \pi I}{r^4} \left(\frac{L}{T^2} \right)$$

C- To find the Moment of Inertia by using two equal masses symmetrically placed on the disc

When two equal masses placed symmetrically placed at a distance d_1 from the axis of the wire at diametrically opposite points

$$I_1 = I + 2I_0 + 2m d_1^2$$

At distance d_2 , $I_2 = I + 2I_0 + 2m d_2^2$

Also, $T_1 = 2\pi \sqrt{\frac{I_1}{C}}$ and $T_2 = 2\pi \sqrt{\frac{I_2}{C}}$

$$T_2^2 - T_1^2 = \frac{(I_2 - I_1)}{I} T^2$$

$$\therefore I = 2m(d_2^2 - d_1^2) \frac{T^2}{T_2^2 - T_1^2}$$

Procedure:

- The length (L) of the wire between the two chucks is adjusted to be, say 0.4 metre. A reference line is drawn on the disc.
- The disc is slightly rotated in the horizontal plane and let free. It executes torsional oscillations. Time for 20 oscillations is determined twice and the period T is calculated. Hence (L/T^2) is calculated.
- The experiment is repeated changing L in steps of 10 cm. (L/T^2) is found to be a constant. Its mean value is taken.
- The radius (r) of the wire is accurately measured using screw gauge.
- Now place the two masses (0.2 kg) at distance d_1 from the axis of the wire at diametrically opposite points. Find the time period T_1 . Change d_1 to d_2 and find T_2 .

Observations:

Mass of the Disc = M=

Radius of the Disc = R=

To find the radius (r) of the wire using screw gauge:-

Value one Pitch scale division = mm

Distance moved by the screw for 5 rotations= mm

Pitch of the screw = $\frac{\text{Distance moved}}{\text{No. of rotations}} =$

No of divisions on the head scale = 100

Least count = L.C.= $\frac{\text{Pitch}}{100} =$

Zero Correction = Z=

S.No	P.S.R	H.S.R	Total= P.S.R+H.S.R x L.C	Corrected Reading = Total + Z

To Determine L/T^2 :

Trial No	Length L	Time for 20 Oscillations			Period T	L/T^2
		1	2	Mean		
1						
2						
3						
4						
5						

To find the Moment of Inertia by Direct Method:

$$I = \frac{MR^2}{2} =$$

To find the Moment of Inertia by using two equal masses symmetrically placed on the disc

Magnitude of equal masses = m= 0.2 kg

Distances d_1 =..... d_2 =.....

L=

	1	2	Mean
T			
T₁			
T₂			

$$I = 2m(d_2^2 - d_1^2) \frac{T^2}{T_2^2 - T_1^2} =$$

Results:

1. Rigidity modulus of the material of the wire of the torsion pendulum=
2. Moment of inertia of the disc of the torsion pendulum
using two identical masses on the disc =
3. Moment of inertia of the disc of the torsion pendulum
by direct method =